

# Dimensional Model Specification (Star Schema)

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## Overview

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This specification describes the OLAP Star Schema for the greenfield\_mp project. The diagram is stored in `docs/diagrams/olap_star_schema.mmd` and renders on GitHub using Mermaid syntax.

## Star Schema Structure

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### Fact Table: `fact_performance_reviews`

- **Grain:** One performance review per row
- **Primary Key:** `review_key` (BIGINT, AUTO\_INCREMENT)
- **Foreign Keys** (all surrogate keys):
  - `employee_key` → `dim_employee.employee_key`
  - `department_key` → `dim_department.department_key`
  - `project_key` → `dim_project.project_key`
  - `date_key` → `dim_date.date_key`
- **Measure Columns:**
  - `review_id` (INT, degenerate dimension - OLTP id)
  - `performance_rating` (TINYINT, 1-4)
  - `job_satisfaction` (TINYINT, 1-4)
  - `environment_satisfaction` (TINYINT, 1-4)
  - `work_life_balance` (TINYINT, 1-4)
  - `percent_salary_hike` (TINYINT, 0-50)
  - `review_score` (DECIMAL(5,2), computed)

## Dimension Tables

### 1. `dim_employee` (SCD Type 2)

- **Primary Key:** `employee_key` (INT, AUTO\_INCREMENT)
- **Business Key:** `employee_id` (INT, from OLTP)
- **Type 2 History Attributes** (tracked over time):
  - `full_name` (VARCHAR(130))
  - `department_name` (VARCHAR(60), SCD Type 2)
  - `job_role` (VARCHAR(60), SCD Type 2)
  - `job_level` (TINYINT, SCD Type 2)
  - `monthly_income` (INT, SCD Type 2)

- `start_date` (DATE, version start)
- `end_date` (DATE, version end, 9999-12-31 for current)
- `is_current` (TINYINT(1), 1 = current version)
- `change_reason` (VARCHAR(40), nullable)
- **Type 1 Attributes** (overwritten, no history):
- `age` (TINYINT UNSIGNED)
- `marital_status` (VARCHAR(15))
- `education_level` (TINYINT)
- `education_field` (VARCHAR(40))
- `gender` (VARCHAR(10))
- `hire_date` (DATE)
- `attrition_flag` (TINYINT(1))
- `over_time_flag` (TINYINT(1))
- **Unique Key:** (employee\_id, start\_date)
- **Index:** (is\_current, department\_name)

## 2. `dim_department`

- **Primary Key:** `department_key` (INT, AUTO\_INCREMENT)
- **Business Key:** `department_id` (INT, from OLTP)
- **Attributes:**
- `department_name` (VARCHAR(60), UNIQUE)
- **Constraints:**
- UNIQUE on `department_id`
- UNIQUE on `department_name`

## 3. `dim_project`

- **Primary Key:** `project_key` (INT, AUTO\_INCREMENT)
- **Business Key:** `project_id` (INT, from OLTP)
- **Attributes:**
- `project_name` (VARCHAR(100))
- `department_name` (VARCHAR(60), nullable)
- `status` (VARCHAR(20), nullable)
- `priority` (VARCHAR(10), nullable)
- `budget` (DECIMAL(12,2), nullable)
- `start_date` (DATE, nullable)
- `end_date` (DATE, nullable)
- **Special Row:** Key -1, "No project" (for reviews without a project)
- **Constraints:**
- UNIQUE on `project_id`

#### 4. `dim_date` (Smart Date Key)

- **Primary Key:** `date_key` (INT, yyyyymmdd format)
- **Attributes:**
  - `full_date` (DATE, UNIQUE)
  - `day_of_month` (TINYINT)
  - `day_name` (VARCHAR(10))
  - `month_number` (TINYINT)
  - `month_name` (VARCHAR(10))
  - `quarter_number` (TINYINT)
  - `year_number` (SMALLINT)
  - `is_weekend` (TINYINT(1), 1 = Saturday/Sunday)
- **Constraints:**
  - UNIQUE on `full_date`
- **Key Generation:** `YEAR(full_date) * 10000 + MONTH(full_date) * 100 + DAY(full_date)`

## Data Flow Diagram

```
IBM HR CSV (1,470 rows)
  |
  v
scripts/synthesize_data.py (pandas + Faker)
  |
  v
data/synthetic/*.csv (100k+ employees, history, projects, reviews)
  |
  v
scripts/load_staging.py
  |
  v
employee_staging (raw, messy)
  |
  v
sp_load_oltp_from_staging (stored procedure)
  |
  v
employee_oltp (normalized OLTP DB)
  |           ^
  |           |
  |           +-- writes (Streamlit app data entry)
  |
  v
sp_run_full_etl (stored procedure)
  |
  v
employee_dw (star schema warehouse)
  |
  v
AnalyticsManager (read-only queries)
  |
  v
Streamlit app dashboards
```

# Draw.io Editing Instructions

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## To Edit in draw.io:

1. Open draw.io (diagrams.net)
2. Use **Arrange > Insert > Advanced > Mermaid** to insert the Mermaid diagram
3. Paste the Mermaid code from `docs/diagrams/olap_star_schema.mmd`
4. The diagram will render automatically
5. To modify, edit the Mermaid code directly
6. Use **Arrange > Layout > Horizontal/Vertical** to adjust layout if needed
7. Save the file and it will update on GitHub

## Current Diagram (olap\_star\_schema.mmd)

```
erDiagram
    DIM_EMPLOYEE ||--o{ FACT_PERFORMANCE_REVIEWS : employee_key
    DIM_DEPARTMENT ||--o{ FACT_PERFORMANCE_REVIEWS : department_key
    DIM_PROJECT ||--o{ FACT_PERFORMANCE_REVIEWS : project_key
    DIM_DATE ||--o{ FACT_PERFORMANCE_REVIEWS : date_key

    FACT_PERFORMANCE_REVIEWS {
        bigint review_key PK
        int review_id "degenerate dimension"
        int employee_key FK
        int department_key FK
        int project_key FK
        int date_key FK
        int performance_rating
        int job_satisfaction
        int environment_satisfaction
        int work_life_balance
        int percent_salary_hike
        decimal review_score
    }
    DIM_EMPLOYEE {
        int employee_key PK "surrogate"
        int employee_id "business key"
        string full_name
        string department_name "SCD2"
        string job_role "SCD2"
        int job_level "SCD2"
        int monthly_income "SCD2"
        date start_date
        date end_date
        bool is_current
    }
    DIM_DEPARTMENT {
        int department_key PK
        int department_id
        string department_name
    }
    DIM_PROJECT {
        int project_key PK
        int project_id
        string project_name
        string status
        string priority
    }
    DIM_DATE {
        int date_key PK
        date full_date
        int month_number
        int quarter_number
        int year_number
        bool is_weekend
    }
```

## SCD Type 2 Design Details

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### Department/Role/Level/Salary Changes

When any of these attributes change: 1. Close the current row: set `end_date` to yesterday, `is_current` = 0 2. Insert new row: `start_date` = today, `end_date` = 9999-12-31, `is_current` = 1 3. All reviews keep referencing the version that was active on their review date

### Type 1 Attributes (Overwritten)

- Age, marital status, attrition, and name are overwritten in place
- No history tracking needed
- Warehouse always has the latest value

### Unknown Member Pattern

- `dim_project` includes a row with key -1: ("No project", "n/a")
- Reviews without a project assignment use key -1 instead of NULL
- Simplifies fact table foreign key constraints

## Query Performance Considerations

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### Index Recommendations

1. `dim_employee(employee_id)` - frequent lookups by employee
2. `dim_employee(is_current, department_name)` - dashboard filters
3. `fact_performance_reviews(date_key, employee_key)` - time-series queries
4. `fact_performance_reviews(department_key)` - department analytics
5. `dim_date(full_date)` - date dimension unique constraint

### Common Query Patterns

1. **Year-over-year trends:** Join fact → dim\_date → group by year
2. **Department performance:** Join fact → dim\_department → aggregate
3. **Employee history:** Query dim\_employee by employee\_id, order by start\_date
4. **Top performers:** Join fact → dim\_employee → DENSE\_RANK by department
5. **Attrition analysis:** Filter dim\_employee where is\_current=1 and attrition\_flag=1